FIGURE — FULL BLEED, NO CAPTION, NO LOGOS

the prettiest super-resolved galaxy we can make.
say nothing. let the room want it.

That image contains no more information about that galaxy than the blurry one it came from.

That is not a criticism.
It is a theorem.

Shannon, 1948. Processing cannot add information about the source.

so the sharpness came from somewhere else. the next twelve minutes are about where.

HOW THIS TALK WORKS

Everything, three times.

A NAME AND A YEAR

the statistics. **None of it is mine and none of it is new.** I will give you the citation and move on.

THE SKY

my group's data. Galaxies, spectra, ten bands. **This is where the surprises are.**

THE ROOM

the same theorem, about people. **No equations.** If you are not a scientist, this column is the whole talk.

seven ideas: distance · gradient · optimization · entropy · divergence · information · the question.

I

Descent

how you move — distance · gradient · optimization

ONE · DISTANCE

What does “closer” mean?

MAHALANOBIS · 1936

There is no distance until you choose one. Euclidean distance is the choice that **every direction matters equally**.

Nothing is ever like that.

THREE METRICS, ONE IMAGE

pixel MSE — what the network minimized.
ellipticity — what lensing needs, to $\mathbf{10^{-3}}$.
line ratios — the actual physics.

It was closest in the only metric nobody needed.

THE ROOM

You decided what counts as *we're fine* before you ever checked.

Then you kept checking that.

resolution decides which differences are visible. a loss decides which visible differences matter.

TWO · GRADIENT

Which way is downhill?

CAUCHY · 1847

Gradient descent is older than the light bulb.

Its entire content is that it is **local**. It knows the point you are standing on, and nothing else.

CHAINS OF LOCAL STEPS

The distance ladder: parallax → Cepheids → supernovae. Every rung locally valid; the chain answerable to nothing.

JAISP's pretext loss is a gradient chosen **because the target's gradient does not exist yet**.

THE ROOM

Nobody chooses a bad year.

They take **three hundred and sixty-five locally reasonable steps**.

a gradient is a first-order statement. nothing checks the composition.

THREE · OPTIMIZATION

Go. Then what?

GALTON · 1886

Regression to the mean. An L2-optimal estimator **is** the posterior mean — and the posterior mean is **shrunk**.

Not a flaw. A definition.

SHAPE MEASUREMENT

Best network: **25–35% ellipticity shrinkage**. Lensing needs 10^{-3}.

That is Galton, 1886, running inside a neural network in 2026.

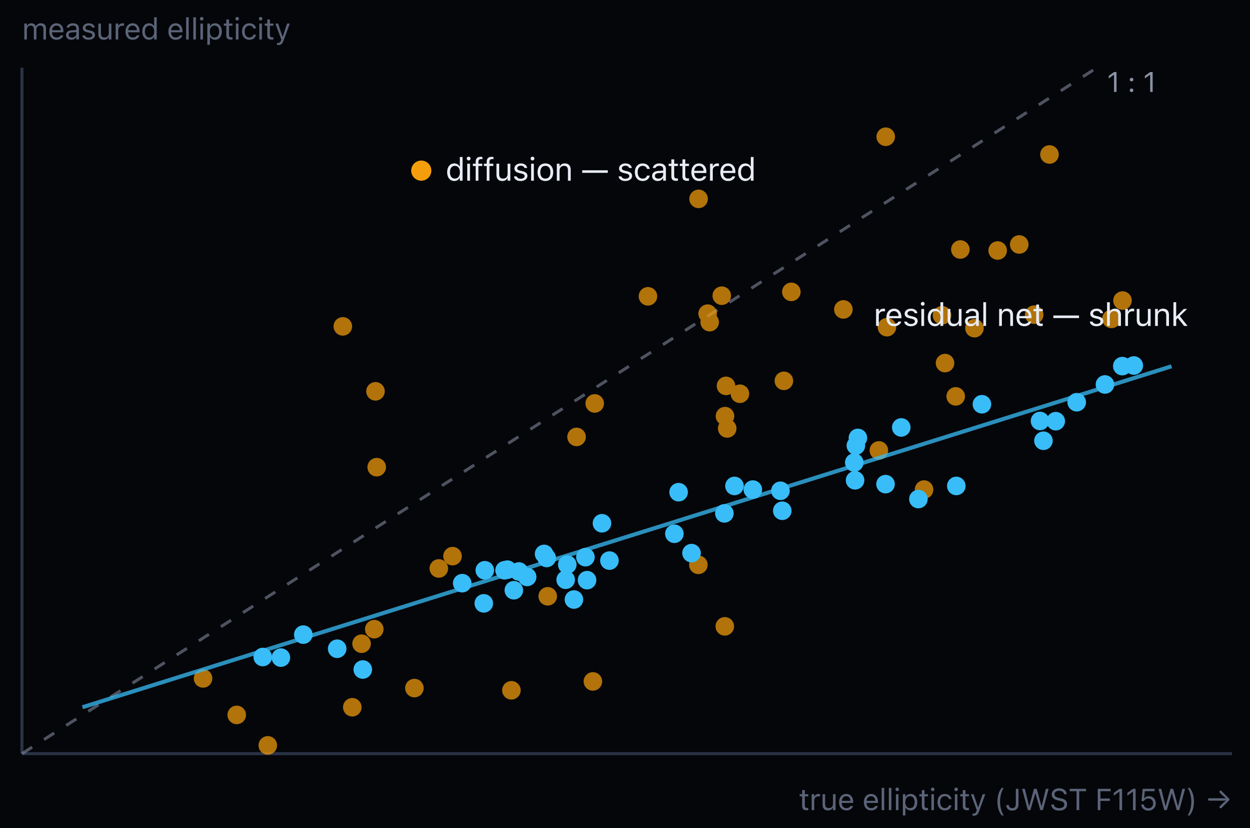
THE ROOM

You fill someone in with **the average of everyone you have known**. That is a posterior mean.

The safest available guess, and guaranteed wrong about the actual person.

the middle column and the right column are the same equation. that is not a metaphor — it is the same estimator.

Two ways to optimize perfectly.



schematic — replace with the paper's scatter

Averaging the posterior gives you low variance and a bias toward the population. Every galaxy drifts round.

Sampling the posterior gives you the right distribution and the *wrong object*.

Which is why diffusion scored below bilinear on shape information.

it isn't lying more than the residual net. it is lying more convincingly.

EVERETTS · HAGHJOO · REZAEE

“Sharper at What Cost?”

every one of these won its own loss. each was audited against a second instrument.

SHAPE · NISP → JWST

diffusion: **below bilinear** on shape info.
best net: **–25–35%** ellipticity.
lensing needs 10^{-3}.

SPECTRA · JADES, 1,187 PAIRS

beat 7 classical methods: **30% lower MAE**, best FWHM.
median Hα recovered: **25.8% of true**.
line ratios — the physics — **degraded**.

PHOTOMETRY · WISE → SPITZER

blended peaks at 3–5″: **35% vs 9%**.
the one win.
and a named bias: oversmoothing pushes flux **up**.

> Learned models are safe where they make a decision you can audit — and dangerous where they make a measurement that propagates in silence.

II

Information

what you actually have — Shannon · Kullback–Leibler · Fisher

FOUR · ENTROPY

How much is actually there?

SHANNON · 1948

Sky → telescope → pixels → network is a chain. **No link can add information about the source.**

The data processing inequality. Not a heuristic — a theorem.

SO WHAT GOT ADDED?

The training set. Information about the population, not about this object.

The audit measured the trade: apparent sharpness up, shape information down.

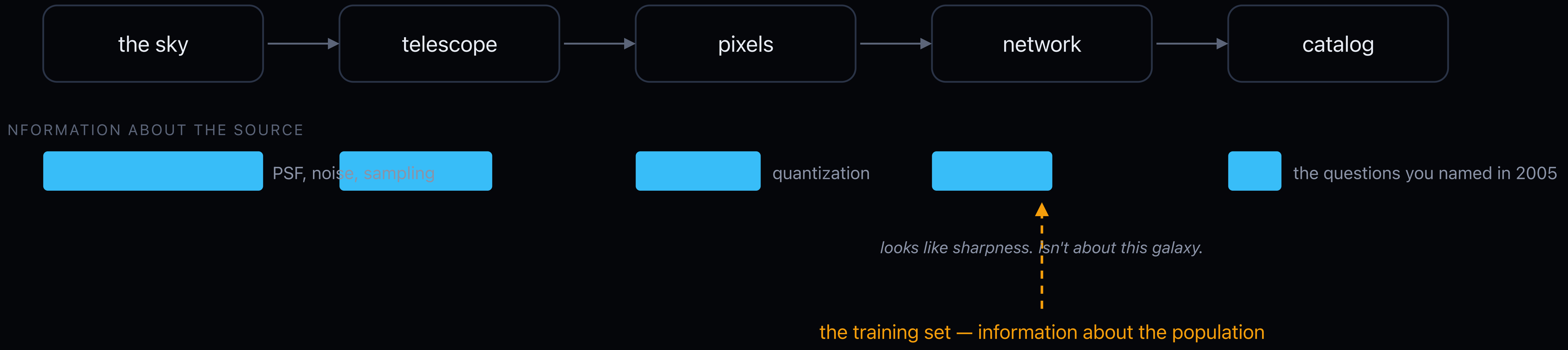
THE ROOM

You cannot learn anything new about someone by thinking about them.

Reflection reorganizes what you already had — usually into something more confident. You have to go and ask.

EVERY ARROW IS A CHANNEL

Where the information went.



the bar can never grow. everything in Act I was a measurement of one of these drops.

FIVE · DIVERGENCE

How far did I actually move?

KULLBACK & LEIBLER · 1951

Not a distance. **Asymmetric** — and the asymmetry is the content: it blows up where reality has mass and your model assigned **none**.

It is also the only honest measure of updating. Entropy changes when you change units. This doesn't.

SURPRISE IS NOT INFORMATION

A noisy pixel is maximally surprising and worthless.

26% of Hα was surprising, and the model was confident. That difference is the whole discipline.

THE ROOM

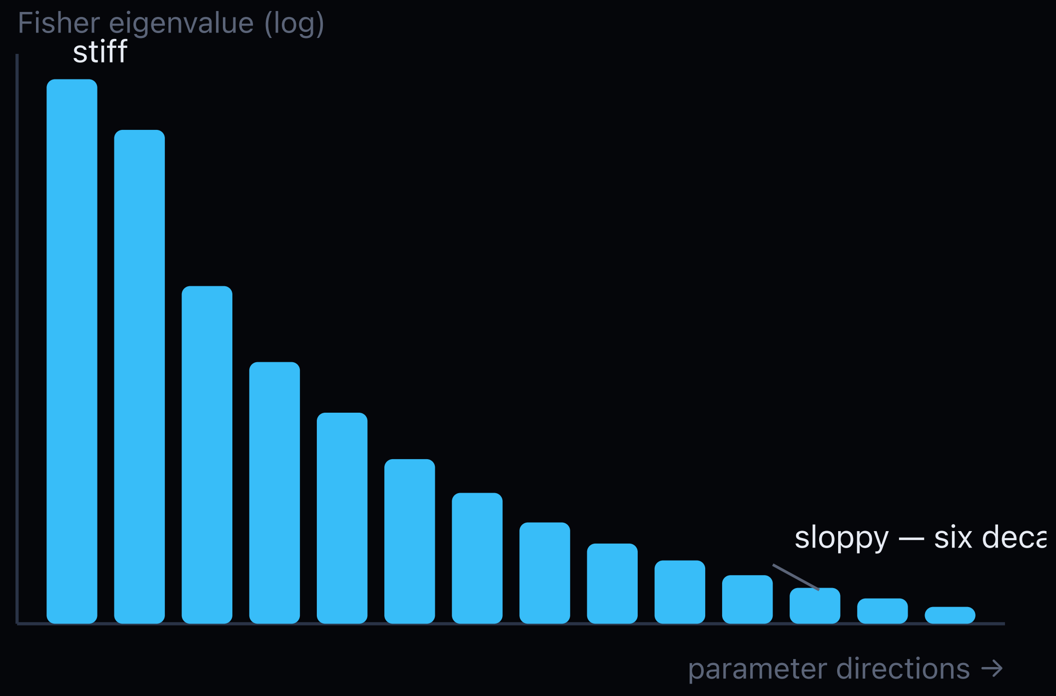
The damage is concentrated entirely in what you ruled out.

Nothing you merely underestimated will cost you much.

every classifier any of us has trained was minimizing a KL. we have all been computing this for years without saying the word.

SIX · IN WHAT SPACE, AND WHICH DIRECTIONS

Which parameters matter?



Fisher · 1922. Locally, KL *is* the Fisher form — so the metric on parameter space is Fisher–Rao.

Chentsov · 1972. It is the *unique* metric invariant under sufficient statistics. There is exactly one right answer to "in what space," and it has been settled for fifty years.

Eigenvalues span five or six decades. A couple of stiff directions, a long sloppy tail. **Occam's razor, derived rather than assumed.**

the room: most of what you argue about is a sloppy direction. two or three are stiff, and they are usually not the ones being discussed.

RAO · 1945 — THE SHARPEST THING ON THIS SLIDE DECK

There is a floor.

The variance of your ellipticity measurement is bounded below by the Fisher information *in the data*. Not in the image you made from it.

A network *can* report error bars under that floor. Only by being biased — and the bias is the training set.

> If your enhanced image reports uncertainties below Cramér–Rao, the error bars are the hallucination — not the pixels.

this is the one thing in the talk you can use tomorrow morning.

SO WHERE DO YOU GET MORE?

You cannot add it.
You can only go and measure again.

Rubin *ugrizy* + Euclid *VIS* · *Y* · *J* · *H* — one sky, ten bands, nobody sees all of it.
Which measurement do you keep, when you do not know next decade's question?

COMMIT

The pipeline writes today's loss in full: **petabytes → kilobytes**. Today's questions, decided in advance, for every source, forever. The compression is real. So is the discard.

HEDGE

JAISP: **mask one band, predict it from the other nine** — each at its delivered sampling. No common grid. No target committed.

three kinds of missing information — never observed · observed and discarded · retained but inaccessible to the model you chose.

JAISP · A JOINT RUBIN–EUCLID FOUNDATION MODEL · HEMMATI ET AL.

Reconstruction is secretly alignment.

To predict the tenth band you must keep exactly what the ten bands *share*. What they share is the sky.
An information bottleneck whose relevant variable is the source — and nobody had to write that down as a loss.

0.45 mag
deeper than the VIS catalog
93% complete · 94% pure

50 → 14 mas
astrometric scatter

0×
retraining on a new field — encoder frozen

~9M
parameters
verify before printing

audited in three layers — the output · the frozen representation · a new head. a useful representation is one that makes several important questions cheap to ask.

III

The Question

whether it ends — and whether that matters

SEVEN · THE QUESTION

The posterior narrows either way.

BERK · 1966

If the truth is **outside** your hypothesis space, the posterior converges to the nearest thing inside it — **with shrinking intervals**.

Precise, and wrong. Lindley, 1956: the value of an experiment is computed *inside* the space. Optimal design cannot find what your prior excluded.

H_0

Two ladders. Both converged. Immaculate residuals. **Incompatible answers.**

A century-old measurement whose frontier is now a diagnosis — and neither ladder can settle it from the inside.

THE ROOM

Certainty is not evidence.

It is what happens when you stop asking the questions whose answers you could not predict.

WEINBERG · THE FIRST THREE MINUTES · 1977

> “The more the universe seems comprehensible, the more it also seems pointless.”

> “The effort to understand the universe is one of the very few things that lifts human life a little above the level of farce, and gives it some of the grace of tragedy.”
>
> THREE SENTENCES LATER

Those are not two moods.

A theory that ends the *why* is, by construction, a theory that answers nothing further. He knew that, and he wanted it anyway.

he died in 2021. the ladder kept going.

EVERYTHING I HAVE USED

None of this is new.

1847 Cauchy — the gradient is local
1886 Galton — the estimator shrinks
1922 Fisher — the metric on the space
1936 Mahalanobis — there is no distance until you pick one
1945 Rao — the floor under your error bars

1948 Shannon — processing cannot add
1951 Kullback & Leibler — the cost of ruling out
1956 Lindley — the value of an experiment
1966 Berk — narrow and wrong
1972 Chentsov — and only one metric will do

The newest idea in this talk is fifty-one years old.

Goodhart, 1975. All of it is one `pip install` away from being ignored.

THE LAST DOMAIN

Prediction error converges.
People don't.

SHANNON

You cannot add information about them by thinking. You have to go and ask.

GALTON

Your model of them shrinks toward everyone else you have met.

KULLBACK–LEIBLER

What hurts is the possibility you assigned zero.

You stop being surprised by someone. That is the proxy converging — not the knowledge.

you are seeing me at podium resolution, which I selected carefully.

GALAXY real clump, or the PSF?

NETWORK physics, or the training set?

A FIELD new law, or an unaudited channel?

US you, or my projection?

measurement, or invention.

حيرت

In Attar, bewilderment is the *sixth* valley. Not the first.
It comes after knowledge, and after unity.

You cannot add information.
You can only go and get more.